Selim Mamak

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EDUCATION

University of Illinois Urbana-Champaign, Champaign, IL — August 2023 - May 2027

Bachelor of Science in Computer Engineering, Minor in Business — *Dean's List/James Scholar Honors | GPA: 3.79/4.0*

Relevant Coursework: Senior Design Laboratory, Computer Systems Engineering, Digital Systems Laboratory, Data Structures, AI, Applied Machine Learning, Database Systems, Computer Systems & Programming, Signal Processing, Discrete Structures

WORK EXPERIENCE

cliexa, *AI/ML Engineer Intern*, Denver, CO — May 2026 - August 2026

- Programmed and shipped an Azure market intelligence platform as sole engineer in two weeks, integrating 24 sources through Azure Functions and PostgreSQL at approximately $40/month.
- Architected a production LangGraph pipeline converting clinical transcripts into structured notes across 36+ document types, integrating retrieval over clinical guidelines and parallel processing to reduce latency by 45%.
- Implemented LLM scoring for approximately 360 items weekly, incorporating required citations, structured output validation, prompt injection defenses, and human review.
- Developed a React/TypeScript dashboard with Entra ID authentication, a natural language copilot, and approved action workflows for Jira and HubSpot; added automated testing and visual regression checks.
- Enhanced a .NET clinical agent platform with FastMCP integrations exposing EHR data and patient APIs to AI agents, supporting clinical workflows across four Azure environments.
- Led client demos and translated clinical and product requirements into deployed AI features.

Microsoft, *AI/ML Engineer Intern*, Istanbul, Turkey — June 2025 - August 2025

- Designed and implemented LSTM and GRU models in PyTorch for financial time series forecasting, improving short term prediction accuracy by 15% using momentum, volatility, and rolling window features.
- Engineered Azure ML pipelines for data preprocessing, training, and evaluation, incorporating sliding window sampling, normalization, and reproducible experiment tracking.
- Automated repetitive experimentation tasks with Python on Azure compute, accelerating iteration across LSTM, GRU, and attention architectures.

Ernst & Young, *Consultant Trainee*, Istanbul, Turkey — July 2025 - August 2025

- Analyzed client case studies in EY's FutureReady Data and Analytics program, applying MECE decomposition and root cause analysis to structure complex business problems.
- Applied hypothesis driven frameworks to evaluate business issues and synthesized findings into executive ready presentations supported by data.

Vestel Electronics, *Software Engineering Intern*, Izmir/Manisa, Turkey — May 2024 - June 2024

- Optimized embedded TV software modules with a 6-engineer team, hardening components shipped on production device lines
- Built an internal tool comparing image file changes across the TV software codebase, increasing debugging efficiency by 25%
- Modeled city-level retail expansion over historical sales data, ranking candidate regions for new store openings

LEADERSHIP EXPERIENCE

IEEE-Eta Kappa Nu (HKN) Honor Society, Student Services *Committee Member*, Champaign, IL — January 2025 - Present

- Led review sessions across Computing, Electronics, Analog Signal Processing, and Computer Systems & Programming, supporting 500+ students with exam readiness and course performance

Dale Carnegie Program, *Scholar*, Kansas City, MO — January 2024

- Earned the Highest Award of Achievement through intensive leadership and communication training across interactive classes, group activities, and presentations

SKILLS

Programming & Software: Python, Java, SQL, C/C++, JavaScript/TypeScript, React, Node.js, .NET, FastAPI

Cloud & Data: Azure Functions, Azure ML, PostgreSQL, Docker, Kubernetes, Pandas, NumPy

Development & Testing: Claude Code, Git, Jira, Automated Testing, Visual Regression Testing

AI & ML: PyTorch, TensorFlow, scikit-learn, LangGraph, RAG, MCP, AI Agent Orchestration, LLM Evaluation, OpenAI API

Engineering: System Architecture, API Integration, Data Pipeline Engineering, Process Automation

Certifications: AI-Empowered SAFe® Product Owner/Manager (2026); McKinsey Forward (2026); DALF C1 (Advanced French)